General Notes

Physics Experiments

Independent physical ideas, models, and exploratory notes.

July 14, 2026

Contents

Chapter 1

Template Independent Note

1.1 Seed Idea

Write the core idea here. This file is a placeholder for an independent physics note created from a discussion, a loose thought, or a larger experiment.

1.2 Working Notes

Use this section to develop physical interpretation, assumptions, models, equations, diagrams, examples, and open questions. The goal is not to force the idea into a course sequence, but to make the thought readable enough that it can later be revised, expanded, or moved into a formal lecture-note project.

Remark

This chapter is intentionally independent. Future notes should usually be added as new `chapter_*.tex` files rather than inserted into this template.

1.3 Possible Next Steps

- Identify the physical system and assumptions.
- Add units, dimensional checks, and limiting cases.
- Record any useful diagrams in `figures/independent_notes/`.
- Decide later whether the note belongs in a formal course project.

Chapter 2

Projectile Motion as a Two-Dimensional Model

2.1 Purpose and Physical Assumptions

Projectile motion is one of the simplest situations in which a two-dimensional motion becomes understandable by separating it into two one-dimensional motions. The horizontal motion is uniform, while the vertical motion is uniformly accelerated by gravity.

In this note we derive the standard formulas for an ideal projectile launched from a point with initial speed v_0 at an angle θ above the horizontal. The purpose is not only to list the formulas, but to show where they come from and what assumptions make them valid.

We use the following idealized model.

- The projectile is treated as a point particle.
- Air resistance is neglected.
- The gravitational acceleration is constant and directed downward.
- The Earth is treated locally as flat, so horizontal and vertical axes are meaningful over the distance considered.
- The launch point is chosen as the origin unless stated otherwise.

With the positive y-axis pointing upward, the acceleration vector is

$$\mathbf{a} = (0, -g),$$

where $g > 0$ is the magnitude of gravitational acceleration.

2.2 Geometry of the Launch

Let the initial velocity be

$$\mathbf{v}_0 = (v_{0x}, v_{0y}).$$

If the initial speed is $v_0 = \|\mathbf{v}_0\|$ and the launch angle is θ, then the velocity components are

$$v_{0x} = v_0 \cos\theta, \qquad v_{0y} = v_0 \sin\theta.$$

This is simply the decomposition of a vector into its horizontal and vertical components.

2.3 Equations of Motion from Constant Acceleration

The acceleration equations are

$$\frac{d^2x}{dt^2} = 0, \qquad \frac{d^2y}{dt^2} = -g.$$

We derive the position functions by integrating with respect to time.

For the horizontal coordinate,

$$\frac{d^2x}{dt^2} = 0.$$

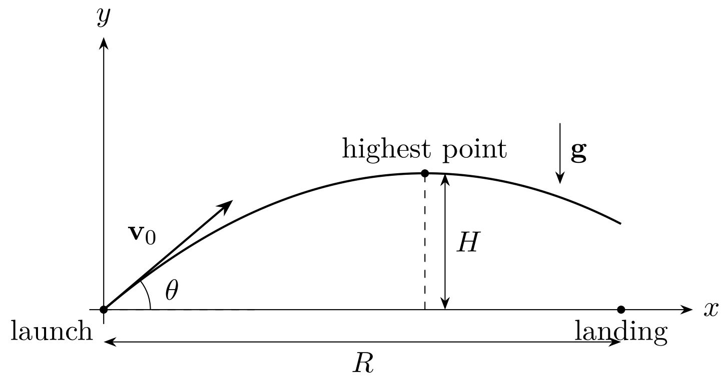

Figure 2.1: Ideal projectile motion without air resistance. The trajectory is a parabola, the acceleration is vertical, and the horizontal and vertical motions can be treated separately.

Integrating once gives

$$\frac{dx}{dt} = C_1.$$

The constant C_1 is the initial horizontal velocity, so

$$\frac{dx}{dt} = v_{0x} = v_0 \cos\theta.$$

Integrating again gives

$$x(t) = v_{0x}t + C_2.$$

If $x(0) = 0$, then $C_2 = 0$, hence

$$\boxed{x(t) = v_0 \cos\theta\, t.}$$

For the vertical coordinate,

$$\frac{d^2y}{dt^2} = -g.$$

Integrating once gives

$$\frac{dy}{dt} = -gt + C_3.$$

The constant C_3 is the initial vertical velocity, so

$$\frac{dy}{dt} = v_{0y} - gt = v_0 \sin\theta - gt.$$

Integrating again gives

$$y(t) = v_{0y}t - \frac{1}{2}gt^2 + C_4.$$

If $y(0) = 0$, then $C_4 = 0$, hence

$$\boxed{y(t) = v_0 \sin\theta\, t - \frac{1}{2}gt^2.}$$

Thus the velocity and position are

$$\boxed{\mathbf{v}(t) = (v_0 \cos\theta,\, v_0 \sin\theta - gt)}$$

and

$$\boxed{\mathbf{r}(t) = (v_0 \cos\theta\, t,\, v_0 \sin\theta\, t - \tfrac{1}{2}gt^2).}$$

Remark

The horizontal coordinate changes linearly in time because there is no horizontal acceleration in the ideal model. The vertical coordinate contains a quadratic term because the vertical acceleration is constant and nonzero.

2.4 The Trajectory as a Parabola

The parametric equations are

$$x(t) = v_0 \cos\theta\, t, \qquad y(t) = v_0 \sin\theta\, t - \frac{1}{2}gt^2.$$

Assume $\cos\theta \neq 0$. From the horizontal equation,

$$t = \frac{x}{v_0 \cos\theta}.$$

Substituting this into $y(t)$ gives

$$y = x\tan\theta - \frac{g}{2v_0^2 \cos^2\theta}x^2.$$

Therefore the trajectory is

$$\boxed{y(x) = x\tan\theta - \frac{g}{2v_0^2 \cos^2\theta}x^2.}$$

This is a quadratic function of x, so the ideal trajectory is a parabola.

2.5 Time of Flight

The projectile returns to the launch height when $y(t) = 0$. With launch and landing at the same height,

$$y(t) = v_0 \sin\theta\, t - \frac{1}{2}gt^2 = 0.$$

Factoring out t,

$$t\left(v_0 \sin\theta - \frac{1}{2}gt\right) = 0.$$

There are two solutions. The first is $t = 0$, the launch moment. The second is found from

$$v_0 \sin\theta - \frac{1}{2}gt = 0.$$

Hence the total time of flight is

$$\boxed{T = \frac{2v_0 \sin\theta}{g}.}$$

This formula applies when the projectile lands at the same vertical level from which it was launched.

2.6 Maximum Height

At the highest point the vertical velocity is zero:

$$v_y(t) = v_0 \sin\theta - gt = 0.$$

Therefore the time to reach the top is

$$\boxed{t_H = \frac{v_0 \sin\theta}{g}.}$$

Substituting t_H into $y(t)$,

$$H = v_0 \sin\theta \left(\frac{v_0 \sin\theta}{g}\right) - \frac{1}{2} g \left(\frac{v_0 \sin\theta}{g}\right)^2.$$

Thus

$$H = \frac{v_0^2 \sin^2\theta}{g} - \frac{v_0^2 \sin^2\theta}{2g} = \frac{v_0^2 \sin^2\theta}{2g}.$$

So the maximum height above the launch point is

$$\boxed{H = \frac{v_0^2 \sin^2\theta}{2g}.}$$

2.7 Horizontal Range

The range R is the horizontal distance traveled during the total time of flight. Since

$$x(t) = v_0 \cos\theta\, t,$$

we have

$$R = x(T) = v_0 \cos\theta\, T.$$

Using $T = 2v_0 \sin\theta / g$,

$$R = v_0 \cos\theta \cdot \frac{2v_0 \sin\theta}{g} = \frac{2v_0^2 \sin\theta \cos\theta}{g}.$$

Since $2 \sin\theta \cos\theta = \sin(2\theta)$,

$$\boxed{R = \frac{v_0^2 \sin(2\theta)}{g}.}$$

The range is maximal when $\sin(2\theta) = 1$, that is when

$$2\theta = 90^\circ, \qquad \theta = 45^\circ.$$

For launch and landing at the same height, the ideal range is therefore maximal at 45°.

2.8 Velocity, Speed, and Direction During Flight

The velocity components are

$$v_x(t) = v_0 \cos\theta, \qquad v_y(t) = v_0 \sin\theta - gt.$$

The speed is the magnitude of the velocity vector:

$$\|\mathbf{v}(t)\| = \sqrt{v_x(t)^2 + v_y(t)^2} = \sqrt{v_0^2 \cos^2\theta + (v_0 \sin\theta - gt)^2}.$$

Thus

$$\boxed{\|\mathbf{v}(t)\| = \sqrt{v_0^2 \cos^2\theta + (v_0 \sin\theta - gt)^2}.}$$

The instantaneous direction angle $\alpha(t)$ of the velocity satisfies

$$\tan\alpha(t) = \frac{v_y(t)}{v_x(t)} = \frac{v_0\sin\theta - gt}{v_0\cos\theta},$$

provided $v_0\cos\theta \neq 0$.

At the highest point, $v_y = 0$, so the velocity is purely horizontal:

$$\mathbf{v}(t_H) = (v_0\cos\theta, 0).$$

2.9 Projectile Launched from an Initial Height

If the projectile is launched from height y_0, the position equations become

$$x(t) = x_0 + v_0\cos\theta\, t,$$

and

$$y(t) = y_0 + v_0\sin\theta\, t - \frac{1}{2}gt^2.$$

If the ground is at $y = 0$, the landing time is obtained from

$$y_0 + v_0\sin\theta\, t - \frac{1}{2}gt^2 = 0.$$

Equivalently,

$$gt^2 - 2v_0\sin\theta\, t - 2y_0 = 0.$$

The physically relevant positive solution is

$$\boxed{T = \frac{v_0\sin\theta + \sqrt{v_0^2\sin^2\theta + 2gy_0}}{g}.}$$

The corresponding range, measured from x_0, is

$$\boxed{R = v_0\cos\theta\, T.}$$

When $y_0 = 0$, the formula reduces to

$$T = \frac{v_0\sin\theta + |v_0\sin\theta|}{g}.$$

For an upward launch with $\sin\theta > 0$, this becomes

$$T = \frac{2v_0\sin\theta}{g},$$

which agrees with the same-height result.

2.10 Energy Check

The same maximum height formula can be checked using conservation of mechanical energy. At launch, the vertical part of the kinetic energy associated with upward motion is

$$\frac{1}{2}m(v_0\sin\theta)^2.$$

At the highest point, this vertical kinetic energy has been converted into gravitational potential energy mgH. Thus

$$\frac{1}{2}mv_0^2\sin^2\theta = mgH.$$

Canceling m and solving for H,

$$H = \frac{v_0^2\sin^2\theta}{2g}.$$

This agrees with the kinematic derivation.

2.11 A Worked Symbolic Example

Suppose a projectile is launched from ground level with speed $v_0 = 20\,\mathrm{m/s}$ at angle $\theta = 30°$. Since

$$\sin 30° = \frac{1}{2}, \qquad \cos 30° = \frac{\sqrt{3}}{2},$$

the initial components are

$$v_{0x} = 10\sqrt{3}\,\mathrm{m/s}, \qquad v_{0y} = 10\,\mathrm{m/s}.$$

The time of flight is

$$T = \frac{2v_{0y}}{g} = \frac{20}{g}.$$

The maximum height is

$$H = \frac{v_{0y}^2}{2g} = \frac{100}{2g} = \frac{50}{g}.$$

The range is

$$R = v_{0x}T = 10\sqrt{3}\cdot\frac{20}{g} = \frac{200\sqrt{3}}{g}.$$

If one later substitutes $g \approx 9.81\,\mathrm{m/s^2}$, these symbolic expressions give numerical predictions in SI units.

2.12 Limitations of the Model

The ideal projectile model is powerful because it is simple, but its assumptions are restrictive. In real motion, air resistance may reduce the range, change the optimal launch angle, and make the trajectory non-parabolic. For high speeds or long distances, the curvature of the Earth and variation of g may also matter. The formulas derived here are therefore best understood as exact results inside an ideal model, and as approximations for real projectiles when the assumptions are sufficiently accurate.

Summary

Projectile motion without air resistance separates into horizontal uniform motion and vertical uniformly accelerated motion. From $\mathbf{a} = (0, -g)$, integration gives

$$x(t) = v_0\cos\theta\, t, \qquad y(t) = v_0\sin\theta\, t - \frac{1}{2}gt^2.$$

Eliminating time gives a parabolic trajectory. For launch and landing at the same height,

$$T = \frac{2v_0\sin\theta}{g}, \qquad H = \frac{v_0^2\sin^2\theta}{2g}, \qquad R = \frac{v_0^2\sin(2\theta)}{g}.$$

Chapter 3

Differential Geometry for General Relativity

3.1 Purpose and Scope

General relativity is not a theory of forces acting in a fixed Euclidean arena. It is a theory in which spacetime itself is represented by a differentiable manifold equipped with a Lorentzian metric. Matter and energy determine the curvature of this metric, while freely falling bodies move along curves selected by the metric connection. For this reason, the mathematical language of general relativity is the language of manifolds, tensor fields, connections, geodesics, and curvature.

This chapter gives a mathematically precise introduction to the differential geometric structures needed before one can state Einstein's field equation in a coordinate-independent way. The discussion is intentionally formal, but each formal object is also connected to its physical role. The point is not to make curvature intuitive by vague analogies. The point is to define the objects correctly and then explain how those objects become physical in relativity.

The guiding chain of ideas is

$$\text{smooth manifold} \longrightarrow \text{tangent spaces} \longrightarrow \text{tensor fields} \longrightarrow \text{metric} \longrightarrow \text{connection} \longrightarrow \text{curvature} \longrightarrow \text{Einstein te}$$

A reader who understands this chain understands the mathematical skeleton of general relativity.

Remark

The word *manifold* should not be interpreted as a surface sitting inside a larger Euclidean space. An abstract manifold is a space which locally has coordinates, but it does not need an ambient space. In general relativity, spacetime is normally treated as an abstract four-dimensional manifold.

3.2 Topological and Smooth Manifolds

A manifold is first a topological object. Smoothness is additional structure. It is important not to confuse the two levels. Topology tells us which points are near which other points; smooth structure tells us which coordinate changes are differentiable.

Definition

An *n-dimensional topological manifold* is a topological space M such that:

1. M is Hausdorff;
2. M is second countable;
3. every point $p \in M$ has an open neighbourhood $U \subseteq M$ homeomorphic to an open subset of $\mathbb{R}^n$.

A pair (U, φ), where $U \subseteq M$ is open and $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ is a homeomorphism, is called an *coordinate chart*.

If $p \in U$, the coordinate map gives

$$\varphi(p) = (x^1(p), \ldots, x^n(p)).$$

The functions $x^1, \ldots, x^n$ are the local coordinate functions. They are not intrinsic physical quantities. They are labels assigned to events or points by a chosen chart.

Two charts (U, φ) and (V, ψ) overlap on $U \cap V$. On the overlap, the coordinate transition map is

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V).$$

This map expresses the new coordinates as functions of the old coordinates. The ability to differentiate tensor fields depends on the differentiability of these transition maps.

Definition

A *smooth atlas* on M is a collection of coordinate charts $\{(U_\alpha, \varphi_\alpha)\}$ covering M such that every transition map

$$\varphi_\beta \circ \varphi_\alpha^{-1}$$

is a C^∞ diffeomorphism between open subsets of $\mathbb{R}^n$. A *smooth manifold* is a topological manifold together with a maximal smooth atlas.

In relativity, the smooth structure allows us to write differentiable fields: worldlines, vector fields, tensor fields, metrics, connections, and curvature. Without this structure, the expression $\partial g_{\mu\nu}/\partial x^\rho$ would have no invariant meaning.

3.3 Coordinate Independence and Diffeomorphisms

The central mathematical discipline in differential geometry is to distinguish objects from their coordinate representations. A physical spacetime event is not an ordered quadruple of numbers. A coordinate system assigns such quadruples to events, but a different coordinate system assigns different quadruples to the same events.

Definition

Let M and N be smooth manifolds. A map $F : M \to N$ is *smooth* if, for every pair of charts (U, φ) on M and (V, ψ) on N with $F(U) \subseteq V$, the coordinate representation

$$\psi \circ F \circ \varphi^{-1}$$

is a smooth map between open subsets of Euclidean spaces.

A diffeomorphism is a smooth bijection with smooth inverse. If $F : M \to N$ is a diffeomorphism, then M and N have the same smooth structure for all differential-geometric purposes.

In general relativity, diffeomorphisms have a special conceptual role. They express the fact that the theory is not tied to a preferred coordinate system. The field equations should be equations between geometric objects on a manifold, not equations whose meaning depends on one selected coordinate grid.

Remark

Coordinate covariance alone is not the same as physical insight. One can write bad physics in a covariant notation. The serious point is that the fundamental fields in general relativity are geometric objects: the metric, curvature, stress-energy tensor, and the maps or curves describing matter.

If x^μ and $x^{\mu'}$ are two coordinate systems on an overlap, then a coordinate transformation is written locally as

$$x^{\mu'} = x^{\mu'}(x^1, \dots, x^n).$$

The Jacobian matrix is

$$\frac{\partial x^{\mu'}}{\partial x^\nu}.$$

The transformation rules for vectors, covectors, and tensors are built from this Jacobian and its inverse. This is why tensor notation is the correct notation for coordinate-independent physics.

3.4 Tangent Vectors as Derivations

In Euclidean space, a tangent vector can be drawn as an arrow. On an abstract manifold there is no ambient vector space in which such an arrow naturally lives. Therefore the tangent vector must be defined intrinsically.

Definition

Let M be a smooth manifold and let $p \in M$. A *tangent vector at* p is a linear map

$$X_p : C^\infty(M) \to \mathbb{R}$$

satisfying the Leibniz rule

$$X_p(fh) = f(p)X_p(h) + h(p)X_p(f)$$

for all smooth functions $f, h \in C^\infty(M)$. The vector space of tangent vectors at p is denoted by T_pM.

This definition encodes directional differentiation. A tangent vector is an operator which differentiates functions at a point.

In local coordinates $(x^1, \ldots, x^n)$, the coordinate tangent vectors are

$$\left.\frac{\partial}{\partial x^1}\right|_p, \ldots, \left.\frac{\partial}{\partial x^n}\right|_p.$$

Every tangent vector can be written uniquely as

$$X_p = X^\mu \left.\frac{\partial}{\partial x^\mu}\right|_p,$$

where the Einstein summation convention is used: repeated upper and lower indices are summed.

If coordinates are changed from x^μ to $x^{\mu'}$, the coordinate basis transforms by the chain rule:

$$\frac{\partial}{\partial x^{\mu'}} = \frac{\partial x^\nu}{\partial x^{\mu'}} \frac{\partial}{\partial x^\nu}.$$

Therefore the components of a vector transform as

$$X^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\nu} X^\nu.$$

The vector itself is invariant; only its coordinate components change.

3.5 Cotangent Vectors and Differentials

The dual space of the tangent space is the cotangent space.

Definition

The *cotangent space* at $p \in M$ is

$$T_p^*M = \mathrm{Hom}(T_pM, \mathbb{R}).$$

Its elements are called *cotangent vectors*, *covectors*, or *one-forms* at p.

If $(x^1, \ldots, x^n)$ is a coordinate system, the dual basis of T_p^*M is

$$dx^1|_p, \ldots, dx^n|_p,$$

where

$$dx^\mu \left(\frac{\partial}{\partial x^\nu} \right) = \delta^\mu_\nu.$$

Here δ^μ_ν is the Kronecker delta.

For a smooth function f, its differential at p is the covector

$$df_p : T_pM \to \mathbb{R}$$

defined by

$$df_p(X_p) = X_p(f).$$

In coordinates,

$$df = \frac{\partial f}{\partial x^\mu} dx^\mu.$$

Thus gradients in differential geometry are first naturally covectors, not vectors. A metric is needed to convert a covector into a vector.

Under coordinate transformation, a covector $\omega = \omega_\mu dx^\mu$ has components transforming as

$$\omega_{\mu'} = \frac{\partial x^\nu}{\partial x^{\mu'}} \omega_\nu.$$

This is the inverse transformation compared with vector components. This is why upper and lower indices have different transformation laws.

3.6 Tensor Spaces

A tensor at a point is a multilinear map built from tangent and cotangent spaces. Tensors are the natural algebraic objects of coordinate-independent physics.

Definition

A tensor of type (r, s) at $p \in M$ is an element of

$$T^r_s(T_pM) = \underbrace{T_pM \otimes \cdots \otimes T_pM}_{r \text{ factors}} \otimes \underbrace{T^*_pM \otimes \cdots \otimes T^*_pM}_{s \text{ factors}}.$$

Equivalently, it may be viewed as a multilinear map with r covector slots and s vector slots.

A type $(1, 1)$ tensor can be viewed as a linear map

$$A : T_pM \to T_pM.$$

A type $(0, 2)$ tensor is a bilinear form

$$B : T_pM \times T_pM \to \mathbb{R}.$$

A type $(2, 0)$ tensor is a bilinear form on covectors.

In coordinates, a tensor of type (r, s) has components

$$T^{\mu_1 \cdots \mu_r}{}_{\nu_1 \cdots \nu_s}.$$

Under a change of coordinates, each upper index receives a factor $\partial x^{\mu'}/\partial x^{\mu}$, and each lower index receives a factor $\partial x^{\nu}/\partial x^{\nu'}$. For example, a type $(1, 1)$ tensor transforms as

$$A^{\mu'}{}_{\nu'} = \frac{\partial x^{\mu'}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x^{\nu'}} A^{\alpha}{}_{\beta}.$$

This transformation law is not a convention. It is what makes the tensor an intrinsic object independent of coordinates.

3.7 Tensor Fields

A tensor field assigns a tensor smoothly to each point of a manifold. Physical fields in relativity are tensor fields on spacetime.

Definition

A *tensor field of type* (r, s) on a smooth manifold M is a smooth assignment

$$p \mapsto T_p \in T^r_s(T_pM).$$

The vector space of smooth type (r, s) tensor fields is often denoted by $\mathcal{T}^r_s(M)$.

Examples include:

$$X = X^\mu \frac{\partial}{\partial x^\mu}$$

for a vector field,

$$\omega = \omega_\mu dx^\mu$$

for a one-form, and

$$T = T_{\mu\nu} dx^\mu \otimes dx^\nu$$

for a covariant two-tensor field.

In relativity, the stress-energy tensor $T_{\mu\nu}$ is a symmetric $(0, 2)$-tensor field. The metric $g_{\mu\nu}$ is also a symmetric $(0, 2)$-tensor field, but with a special nondegeneracy and signature property. Curvature tensors are built from the metric and its associated connection.

A coordinate expression such as T_{00} is not by itself a scalar physical quantity. It is a component relative to a chosen coordinate basis. Meaningful statements must be made either tensorially or by specifying the observer or frame relative to which the component is measured.

3.8 Vector Fields and Integral Curves

A vector field may be understood as an infinitesimal flow on a manifold. At each point it gives a direction and rate of change.

Definition

A *vector field* on M is a smooth section of the tangent bundle TM. In local coordinates it has the form

$$X = X^\mu \frac{\partial}{\partial x^\mu},$$

where the component functions X^μ are smooth.

An integral curve of X is a smooth curve $\gamma : I \to M$ satisfying

$$\dot{\gamma}(t) = X_{\gamma(t)}.$$

In coordinates this becomes the system of ordinary differential equations

$$\frac{dx^\mu}{dt} = X^\mu(x^1(t), \ldots, x^n(t)).$$

The existence and uniqueness theorem for ordinary differential equations implies local existence and uniqueness of integral curves for smooth vector fields.

Theorem

Let X be a smooth vector field on M, and let $p \in M$. There exists an open interval $(-\varepsilon, \varepsilon)$ and a unique integral curve γ with $\gamma(0) = p$, defined at least for sufficiently small time.

Physically, a vector field can represent a velocity field of a continuous medium, a timelike observer congruence, or a generator of a symmetry. In relativity, a timelike vector field may represent a family of observers, while a Killing vector field represents an infinitesimal spacetime symmetry.

3.9 Lie Brackets and Noncommuting Directions

Vector fields act on smooth functions as derivations. Given two vector fields X and Y, their commutator measures the failure of differentiating first along X and then along Y to equal differentiating in the opposite order.

Definition

The *Lie bracket* of two vector fields X and Y is the vector field $[X, Y]$ defined by

$$[X, Y](f) = X(Y(f)) - Y(X(f))$$

for every smooth function f.

If

$$X = X^\mu \partial_\mu, \qquad Y = Y^\mu \partial_\mu,$$

then

$$[X, Y]^\mu = X^\nu \partial_\nu Y^\mu - Y^\nu \partial_\nu X^\mu.$$

The Lie bracket is bilinear, antisymmetric, and satisfies the Jacobi identity

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

Coordinate vector fields commute:

$$\left[\frac{\partial}{\partial x^\mu}, \frac{\partial}{\partial x^\nu}\right] = 0.$$

A general frame need not commute. Noncommuting frames are common in physics, especially in orthonormal tetrad formalisms, rotating frames, and gauge-theoretic formulations.

The Lie bracket is defined without a metric and without a connection. It belongs to the differentiable structure of the manifold.

3.10 Differential Forms

Differential forms are antisymmetric covariant tensor fields. They provide the natural language for integration on manifolds, orientation, volume forms, and field strengths.

Definition

A *differential k-form* on M is a smooth section of $\Lambda^k T^*M$. Equivalently, it is an antisymmetric covariant tensor field of rank k.

A one-form is written

$$\alpha = \alpha_\mu dx^\mu.$$

A two-form is written

$$\omega = \frac{1}{2}\omega_{\mu\nu} dx^\mu \wedge dx^\nu,$$

where $\omega_{\mu\nu} = -\omega_{\nu\mu}$. The wedge product is antisymmetric:

$$dx^\mu \wedge dx^\nu = -dx^\nu \wedge dx^\mu.$$

The exterior derivative is a coordinate-independent operator

$$d : \Omega^k(M) \to \Omega^{k+1}(M)$$

satisfying

$$d^2 = 0.$$

For a function f,

$$df = \partial_\mu f \, dx^\mu.$$

For a one-form $\alpha = \alpha_\mu dx^\mu$,

$$d\alpha = \frac{1}{2}(\partial_\mu \alpha_\nu - \partial_\nu \alpha_\mu) dx^\mu \wedge dx^\nu.$$

In electromagnetism, the electromagnetic field strength is naturally a two-form F, and the homogeneous Maxwell equations are simply

$$dF = 0.$$

This example shows why differential forms are not merely mathematical ornament: they are structurally efficient descriptions of physical fields.

3.11 Metrics: Riemannian and Lorentzian

A metric is not merely a device for measuring distances. It is a smoothly varying nondegenerate bilinear form on tangent spaces. Its signature determines the causal character of vectors.

Definition

A *pseudo-Riemannian metric* on a smooth manifold M is a smooth symmetric $(0, 2)$-tensor field g such that, for every $p \in M$, the bilinear form

$$g_p : T_pM \times T_pM \to \mathbb{R}$$

is nondegenerate.

Nondegenerate means that if

$$g_p(X, Y) = 0$$

for all $Y \in T_pM$, then $X = 0$. In coordinates,

$$g = g_{\mu\nu} dx^\mu \otimes dx^\nu,$$

and nondegeneracy means

$$\det(g_{\mu\nu}) \neq 0.$$

A Riemannian metric is positive definite:

$$g_p(X, X) > 0$$

for every nonzero $X \in T_pM$. A Lorentzian metric on a four-dimensional manifold has signature either $(- + ++)$ or $(+ - --)$. In this chapter we use the convention $(- + ++)$.

For Lorentzian signature, a nonzero tangent vector X is:

timelike	if $g(X, X) < 0$,
null or lightlike	if $g(X, X) = 0$,
spacelike	if $g(X, X) > 0$.

This classification is the mathematical origin of causal structure in general relativity.

3.12 Line Element and Proper Time

In coordinates, the metric is often written through the line element

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

This expression is shorthand for the quadratic form determined by the metric. It should not be understood as an equation between infinitesimal numbers in a naive sense.

For a smooth curve $\gamma : \lambda \mapsto x^\mu(\lambda)$, the tangent vector has components

$$\dot{x}^\mu = \frac{dx^\mu}{d\lambda}.$$

The squared norm of the tangent is

$$g(\dot{\gamma}, \dot{\gamma}) = g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}.$$

For a timelike curve in signature $(- + ++)$, proper time satisfies

$$d\tau^2 = -\frac{1}{c^2} ds^2.$$

If units with $c = 1$ are used, then

$$d\tau = \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}.$$

Thus the metric determines the physical time measured by an ideal clock moving along a timelike worldline.

For a null curve,

$$ds^2 = 0.$$

The light cone at a point consists of all null tangent directions. Therefore the metric determines which events can be connected by light signals and which curves can represent massive particles.

3.13 Raising and Lowering Indices

Because the metric is nondegenerate, it defines an isomorphism between tangent and cotangent spaces:

$$\flat : T_pM \to T_p^*M, \qquad X \mapsto X^\flat,$$

where

$$X^\flat(Y) = g(X, Y).$$

In components,

$$X_\mu = g_{\mu\nu}X^\nu.$$

The inverse metric g^{-1} has components $g^{\mu\nu}$ satisfying

$$g^{\mu\alpha}g_{\alpha\nu} = \delta^\mu_\nu.$$

It raises indices:

$$\omega^\mu = g^{\mu\nu}\omega_\nu.$$

This operation is not harmless notation. It uses the metric. Without a metric, there is no canonical way to identify vectors and covectors.

For a rank-two tensor, one may form mixed components such as

$$T^\mu{}_\nu = g^{\mu\alpha}T_{\alpha\nu}.$$

In relativity this is common for the stress-energy tensor, the Ricci tensor, and the Einstein tensor. The metric is therefore not simply another tensor field; it also controls the algebra of index manipulation.

3.14 Volume Element

A metric determines a natural volume density. If $g_{\mu\nu}$ is the matrix of the metric in coordinates, write

$$g = \det(g_{\mu\nu}).$$

For a Lorentzian metric with signature $(- + ++)$, this determinant is negative in standard orientations, so the invariant volume element is written as

$$dV_g = \sqrt{|g|}\, d^n x.$$

In four-dimensional spacetime,

$$dV_g = \sqrt{-g}\, d^4 x$$

when $g < 0$.

This factor is essential in generally covariant action principles. The Einstein-Hilbert action is

$$S_{\mathrm{EH}} = \frac{1}{16\pi G} \int_M R\sqrt{-g}\, d^4 x,$$

where R is the scalar curvature. The factor $\sqrt{-g}$ makes the integral independent of the coordinate system.

If coordinates change from x^μ to $x^{\mu'}$, the coordinate volume form changes by the absolute value of the Jacobian determinant, while $\sqrt{|g|}$ changes by the inverse factor. Their product is invariant.

3.15 Connections and Covariant Derivatives

Ordinary partial derivatives of tensor components do not transform tensorially under general coordinate changes. A connection corrects this failure and defines differentiation of tensor fields along vector fields.

Definition

An *affine connection* on a smooth manifold M is a map

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M), \qquad (X, Y) \mapsto \nabla_X Y,$$

satisfying, for smooth functions f and vector fields X, Y, Z:

1. $\nabla_{fX+Y} Z = f\nabla_X Z + \nabla_Y Z$;
2. $\nabla_X (Y + Z) = \nabla_X Y + \nabla_X Z$;
3. $\nabla_X (fY) = X(f)Y + f\nabla_X Y$.

The connection is linear in the direction argument and satisfies a Leibniz rule in the field being differentiated.

In a coordinate basis,

$$\nabla_{\partial_\mu} \partial_\nu = \Gamma^\rho{}_{\nu\mu} \partial_\rho.$$

Many texts write the lower indices in the order $\Gamma^\rho{}_{\mu\nu}$ using $\nabla_{\partial_\mu} \partial_\nu = \Gamma^\rho{}_{\mu\nu} \partial_\rho$. In this chapter we use the common relativity convention

$$\nabla_\mu V^\rho = \partial_\mu V^\rho + \Gamma^\rho{}_{\mu\nu} V^\nu.$$

The coefficients $\Gamma^\rho{}_{\mu\nu}$ are the Christoffel symbols of the connection in the chosen coordinates.

3.16 Covariant Derivative of Tensor Fields

For a vector field $V = V^\mu \partial_\mu$, the covariant derivative is

$$\nabla_\mu V^\rho = \partial_\mu V^\rho + \Gamma^\rho{}_{\mu\nu} V^\nu.$$

For a one-form $\omega = \omega_\nu dx^\nu$, the covariant derivative is

$$\nabla_\mu \omega_\nu = \partial_\mu \omega_\nu - \Gamma^\rho{}_{\mu\nu} \omega_\rho.$$

The sign is negative because covectors transform oppositely to vectors.

For a type $(1,1)$ tensor $A^\rho{}_\sigma$,

$$\nabla_\mu A^\rho{}_\sigma = \partial_\mu A^\rho{}_\sigma + \Gamma^\rho{}_{\mu\lambda} A^\lambda{}_\sigma - \Gamma^\lambda{}_{\mu\sigma} A^\rho{}_\lambda.$$

The rule is systematic: add one connection term for each upper index and subtract one connection term for each lower index.

The covariant derivative is constructed so that $\nabla_\mu T^{\alpha\cdots}{}_{\beta\ldots}$ transforms as a tensor with one additional lower index. This is the essential reason it replaces partial derivatives in curved spacetime.

Remark

The Christoffel symbols themselves are not tensor components. They can vanish at a chosen point in suitable coordinates, but their derivatives generally cannot all be made to vanish. Curvature is built from combinations of first derivatives and quadratic products of Christoffel symbols that together transform tensorially.

3.17 Torsion

An affine connection has torsion if the antisymmetric part of its action on vector fields differs from the Lie bracket.

Definition

The *torsion tensor* of a connection ∇ is the type $(1,2)$ tensor field

$$T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y].$$

In a coordinate basis, where $[\partial_\mu, \partial_\nu] = 0$, the components are

$$T^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\nu\mu}.$$

The connection is torsion-free precisely when

$$\Gamma^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\nu\mu}$$

in every coordinate chart.

Standard general relativity uses the torsion-free Levi-Civita connection. Some extensions of relativity, such as Einstein-Cartan theory, allow torsion and relate it to spin density. In the standard theory, however, torsion is set to zero, and the gravitational field is represented by curvature of the metric connection.

3.18 Metric Compatibility

A connection is compatible with a metric when parallel transport preserves inner products.

Definition

A connection ∇ is *metric-compatible* with g if

$$\nabla g = 0.$$

In coordinates this means

$$\nabla_\lambda g_{\mu\nu} = 0.$$

Expanding the coordinate expression gives

$$\nabla_\lambda g_{\mu\nu} = \partial_\lambda g_{\mu\nu} - \Gamma^\rho{}_{\lambda\mu} g_{\rho\nu} - \Gamma^\rho{}_{\lambda\nu} g_{\mu\rho}.$$

Thus metric compatibility is

$$\partial_\lambda g_{\mu\nu} = \Gamma^\rho{}_{\lambda\mu} g_{\rho\nu} + \Gamma^\rho{}_{\lambda\nu} g_{\mu\rho}.$$

Geometrically, this condition means that lengths and angles, or in Lorentzian geometry causal inner products, are preserved under parallel transport.

In relativity, metric compatibility means that the connection used to define free fall is consistent with the metric used to define clocks, rulers, and light cones. Without compatibility, these structures would not be tied together in the standard way.

3.19 The Levi-Civita Connection

The central connection in Riemannian and Lorentzian geometry is the unique connection that is both torsion-free and metric-compatible.

Theorem

Let (M, g) be a pseudo-Riemannian manifold. There exists a unique affine connection ∇ such that:

$$T = 0, \qquad \nabla g = 0.$$

This connection is called the *Levi-Civita connection* of g.

In coordinates, its Christoffel symbols are

$$\boxed{\Gamma^{\rho}{}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} \left(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}\right).}$$

This formula follows by writing metric compatibility in three permutations of the indices and combining them while using the symmetry $\Gamma^{\rho}{}_{\mu\nu} = \Gamma^{\rho}{}_{\nu\mu}$.

The Levi-Civita connection is not an additional independent field once the metric is given. In standard general relativity, the metric determines the connection, and the connection determines the curvature. Thus the gravitational field is encoded in the metric.

3.20 Parallel Transport

Given a connection, one can define what it means for a vector to be transported along a curve without changing relative to the connection.

Let $\gamma : I \to M$ be a smooth curve with tangent vector $\dot{\gamma}$. A vector field $V(t)$ along γ is parallel if

$$\nabla_{\dot{\gamma}} V = 0.$$

In coordinates, this becomes

$$\frac{dV^\rho}{dt} + \Gamma^\rho{}_{\mu\nu} \frac{dx^\mu}{dt} V^\nu = 0.$$

This is a system of linear ordinary differential equations along the curve.

Parallel transport depends on the path in a curved manifold. If a vector is parallel transported around a closed loop, it may return rotated or boosted relative to its initial value. This failure of path independence is measured by curvature.

In general relativity, parallel transport formalizes comparison of vectors at different spacetime points. Since tangent spaces at different points are different vector spaces, there is no canonical comparison without additional structure. The connection supplies such a comparison.

3.21 Geodesics

A geodesic is a curve whose tangent vector is parallel transported along itself. It is the curved-space generalization of a straight line with constant velocity.

Definition

A curve $\gamma : I \to M$ is an *affinely parametrized geodesic* of a connection ∇ if

$$\nabla_{\dot{\gamma}}\dot{\gamma} = 0.$$

In coordinates, this equation is

$$\boxed{\frac{d^2x^\rho}{d\lambda^2} + \Gamma^\rho{}_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0.}$$

For the Levi-Civita connection of a Lorentzian metric, timelike geodesics are the worldlines of freely falling massive test particles, while null geodesics are the paths of ideal light rays in geometric optics.

The word *test* is important. A test particle is assumed not to disturb the metric appreciably. The geodesic equation describes motion in a given geometry; the Einstein equation describes how matter determines that geometry.

3.22 Geodesics from a Variational Principle

For a Riemannian metric, geodesics locally extremize length. In Lorentzian geometry, timelike geodesics locally extremize proper time under suitable conditions.

Consider the energy functional

$$E[\gamma] = \frac{1}{2}\int_{\lambda_0}^{\lambda_1} g_{\mu\nu}(x)\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda}\,d\lambda.$$

The Euler-Lagrange equations for the Lagrangian

$$L = \frac{1}{2}g_{\mu\nu}(x)\dot{x}^\mu\dot{x}^\nu$$

are

$$\frac{d}{d\lambda}\left(g_{\rho\nu}\dot{x}^\nu\right) - \frac{1}{2}\partial_\rho g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0.$$

Expanding the derivative gives

$$g_{\rho\nu}\ddot{x}^\nu + \partial_\sigma g_{\rho\nu}\dot{x}^\sigma\dot{x}^\nu - \frac{1}{2}\partial_\rho g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0.$$

Multiplying by $g^{\rho\alpha}$ and symmetrizing the terms in $\dot{x}^\mu\dot{x}^\nu$ yields

$$\ddot{x}^\alpha + \Gamma^\alpha{}_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0.$$

Thus the variational and connection definitions agree for the Levi-Civita connection.

3.23 Normal Coordinates and the Equivalence Principle

At any point of a pseudo-Riemannian manifold, one can choose coordinates in which the metric takes its flat canonical form and the Christoffel symbols of the Levi-Civita connection vanish at that point.

Theorem

Let (M, g) be a pseudo-Riemannian manifold and let $p \in M$. There exist local coordinates around p such that

$$g_{\mu\nu}(p) = \eta_{\mu\nu}, \qquad \Gamma^{\rho}{}_{\mu\nu}(p) = 0,$$

where $\eta_{\mu\nu}$ is the diagonal flat metric with the same signature as g.

These are normal coordinates at p. In Lorentzian geometry they are closely related to local inertial frames.

However, the first derivatives of the connection cannot generally be made to vanish at the same point. The curvature tensor remains. This distinction is physically crucial: gravitational acceleration can be transformed away at a point by going to a freely falling frame, but tidal effects cannot be eliminated if curvature is nonzero.

This is the precise mathematical content behind the local equivalence principle: locally at a point, freely falling coordinates remove the connection coefficients, but they do not remove curvature.

3.24 Curvature as Noncommutation of Covariant Derivatives

Curvature measures the failure of second covariant derivatives to commute, after correcting for the noncommutation of the vector fields themselves.

Definition

The *Riemann curvature tensor* of a connection ∇ is the map

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

For each pair (X,Y), the map $Z \mapsto R(X,Y)Z$ is a linear endomorphism of the tangent space. In coordinates,

$$R(\partial_\mu, \partial_\nu)\partial_\sigma = R^\rho{}_{\sigma\mu\nu}\partial_\rho.$$

With the sign convention used here,

$$\boxed{R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho{}_{\nu\sigma} - \partial_\nu \Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda}\Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda}\Gamma^\lambda{}_{\mu\sigma}.}$$

Different books sometimes use the opposite sign convention for the Riemann tensor. What matters is internal consistency. The definitions of Ricci tensor, scalar curvature, and Einstein tensor must follow the chosen convention.

3.25 Curvature and Parallel Transport Around a Loop

Curvature has a direct geometric interpretation through parallel transport. If a vector is parallel transported around a very small coordinate parallelogram spanned by coordinate displacements Δx^μ and Δx^ν, the change in the vector is, to leading order,

$$\Delta V^\rho \approx R^\rho{}_{\sigma\mu\nu} V^\sigma \Delta x^\mu \Delta x^\nu.$$

Thus curvature measures the infinitesimal holonomy of the connection.

In flat Euclidean space, parallel transport is path-independent. On a curved surface or curved spacetime, path dependence is the rule. The Riemann tensor is the local object that encodes this path dependence.

Physically, this is connected to tidal effects. A uniform gravitational field can be removed locally by choosing an accelerated or freely falling frame, but the relative acceleration of nearby freely falling particles is controlled by curvature and cannot be removed by a coordinate transformation.

3.26 Algebraic Symmetries of the Riemann Tensor

For the Levi-Civita connection of a metric, it is useful to lower the first index:

$$R_{\rho\sigma\mu\nu} = g_{\rho\lambda} R^{\lambda}{}_{\sigma\mu\nu}.$$

The Riemann tensor then satisfies the algebraic symmetries

$$R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu},$$

$$R_{\rho\sigma\mu\nu} = -R_{\rho\sigma\nu\mu},$$

$$R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma},$$

and the first Bianchi identity

$$R_{\rho[\sigma\mu\nu]} = 0.$$

The brackets denote antisymmetrization over the enclosed indices.

These identities drastically reduce the number of independent components. In n dimensions, the Riemann tensor of a Levi-Civita connection has

$$\frac{n^2(n^2-1)}{12}$$

independent components. In four dimensions this gives

$$\frac{16(15)}{12} = 20.$$

Thus four-dimensional spacetime curvature has twenty independent components at a point before field equations impose additional restrictions.

3.27 Ricci Tensor and Scalar Curvature

The Ricci tensor is a contraction of the Riemann tensor. With the convention of this chapter, define

$$\mathrm{Ric}_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}.$$

In index notation this is often written as

$$R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}.$$

The scalar curvature is the trace of the Ricci tensor with the inverse metric:

$$R = g^{\mu\nu} R_{\mu\nu}.$$

The same letter R is commonly used both for the Riemann curvature operator and for the scalar curvature; context and indices determine the meaning.

The Ricci tensor measures part of curvature related to volume distortion and geodesic convergence. The full Riemann tensor contains more information than the Ricci tensor. In four-dimensional general relativity, the vacuum equation $R_{\mu\nu} = 0$ does not imply that the full Riemann tensor is zero. Vacuum spacetimes may still contain gravitational waves or tidal curvature.

The scalar curvature is a single number at each point. It is too compressed to describe all curvature, but it is central in the Einstein-Hilbert action.

3.28 Sectional Curvature and Riemannian Interpretation

In Riemannian geometry, sectional curvature gives the curvature of a two-plane inside a tangent space. Let $u, v \in T_pM$ be linearly independent. The sectional curvature of the plane they span is

$$K(u, v) = \frac{g(R(u, v)v, u)}{g(u, u)g(v, v) - g(u, v)^2}.$$

This expression generalizes Gaussian curvature of surfaces.

For a two-dimensional Riemannian manifold, all curvature information is encoded by Gaussian curvature. For dimensions three and higher, sectional curvatures for all two-planes encode the Riemann tensor.

In Lorentzian geometry, one must be more careful because the denominator may vanish for degenerate two-planes involving null directions. Nevertheless, the conceptual point survives: curvature is not merely a scalar quantity. It depends on directions and planes in tangent space.

In physics this matters because tidal effects depend on directions. Curvature can stretch in one spatial direction and compress in another. The Riemann tensor keeps this directional information.

3.29 Geodesic Deviation

Geodesic deviation gives the most direct physical interpretation of spacetime curvature. Consider a one-parameter family of geodesics with tangent vector U and separation vector ξ. If U is affinely parametrized, the relative acceleration satisfies

$$\boxed{\nabla_U \nabla_U \xi = R(U, \xi)U.}$$

In components, for a timelike geodesic parametrized by proper time τ,

$$\frac{D^2 \xi^\mu}{d\tau^2} = R^\mu{}_{\nu\rho\sigma} U^\nu \xi^\rho U^\sigma,$$

up to the sign convention for R.

This equation states that curvature controls the relative acceleration of nearby freely falling particles. This is not a coordinate artifact. It is an invariant measurement principle: if a cloud of freely falling particles changes shape due to tidal effects, spacetime curvature is being detected.

In a local inertial frame at a point, the Christoffel symbols can be set to zero at that point, but the second relative acceleration encoded by geodesic deviation remains if the Riemann tensor is nonzero.

3.30 The Bianchi Identities

The first Bianchi identity is algebraic:

$$R^{\rho}{}_{[\sigma\mu\nu]} = 0.$$

For the Levi-Civita connection there is also a differential identity, the second Bianchi identity:

$$\nabla_\lambda R^{\rho}{}_{\sigma\mu\nu} + \nabla_\mu R^{\rho}{}_{\sigma\nu\lambda} + \nabla_\nu R^{\rho}{}_{\sigma\lambda\mu} = 0.$$

Contracting the second Bianchi identity yields

$$\nabla^\mu \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) = 0.$$

This contracted identity is one of the structural reasons why the Einstein tensor appears in the field equations.

The Bianchi identities are not optional extra equations. They are identities satisfied by curvature constructed from the Levi-Civita connection. Their physical consequence is connected to local conservation of stress-energy in general relativity.

3.31 The Einstein Tensor

The Einstein tensor is defined by

$$\boxed{G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}.}$$

It is symmetric because the Ricci tensor and metric are symmetric. By the contracted Bianchi identity,

$$\nabla^\mu G_{\mu\nu} = 0.$$

This divergence-free property is essential for coupling geometry to matter.

Einstein's field equation, with cosmological constant, is

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.}$$

Here $T_{\mu\nu}$ is the stress-energy tensor. The left-hand side is geometry; the right-hand side is matter and energy.

The equation is not simply a poetic statement that matter curves spacetime. It is a precise nonlinear system of partial differential equations for the metric components $g_{\mu\nu}$, because the Ricci tensor contains second derivatives of the metric and nonlinear terms in its first derivatives.

3.32 Stress-Energy Tensor and Conservation

The stress-energy tensor represents local energy density, momentum density, energy flux, and stress as measured relative to observers or frames. In a local orthonormal frame, components have the schematic interpretation

$$T_{00} = \text{energy density}, \qquad T_{0i} = \text{momentum or energy flux}, \qquad T_{ij} = \text{stress components}.$$

These interpretations require a frame. The tensor itself is coordinate independent.

The Einstein equation implies

$$\nabla^\mu T_{\mu\nu} = 0$$

provided Λ is constant. This is the covariant local conservation law. It is not the same as ordinary global conservation of total energy in arbitrary curved spacetimes. Global conservation laws require additional structure, such as asymptotic flatness or spacetime symmetries.

For a perfect fluid with energy density ρ, pressure p, and four- velocity u^μ, the stress-energy tensor is

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + pg_{\mu\nu}$$

in units where $c = 1$, with signature $(- + ++)$ and normalization

$$g_{\mu\nu}u^\mu u^\nu = -1.$$

This formula shows how matter models enter the geometric field equation.

3.33 Local Frames and Tetrads

A coordinate basis ∂_μ is not usually orthonormal. In relativity it is often useful to introduce a local frame e_a, where Latin indices label frame components. A tetrad in four dimensions is a basis

$$e_0, e_1, e_2, e_3$$

of each tangent space such that

$$g(e_a, e_b) = \eta_{ab}.$$

Here $\eta_{ab} = \mathrm{diag}(-1, 1, 1, 1)$ in the chosen convention.

The relation between coordinate and frame components is written

$$e_a = e_a{}^\mu \partial_\mu.$$

The metric can be reconstructed from tetrads by

$$g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu.$$

Physically, an orthonormal tetrad can represent the measuring frame of a local observer: one timelike unit vector and three spacelike unit vectors. Tensor components in this frame correspond more directly to measured quantities than arbitrary coordinate components.

Tetrads are also important in coupling spinor fields to gravity, because spinors are naturally associated with local Lorentz frames rather than directly with coordinate frames.

3.34 Killing Vector Fields and Symmetries

A vector field generates an infinitesimal symmetry of the metric if the metric does not change along its flow. This is expressed by the Lie derivative.

Definition

A vector field K is a *Killing vector field* of a metric g if

$$\mathcal{L}_K g = 0.$$

Equivalently, for the Levi-Civita connection,

$$\nabla_\mu K_\nu + \nabla_\nu K_\mu = 0.$$

Killing vector fields encode continuous spacetime symmetries. A timelike Killing field corresponds to stationarity; a rotational Killing field corresponds to rotational symmetry.

Along a geodesic with tangent U, a Killing field gives a conserved quantity:

$$\frac{d}{d\lambda} g(K, U) = 0.$$

This follows from the geodesic equation and the Killing equation. In physical terms, symmetries generate conserved quantities such as energy and angular momentum, when the relevant symmetry exists.

3.35 Flatness and Curved Coordinates

It is important to distinguish nonzero Christoffel symbols from genuine curvature. Curvilinear coordinates in flat space can produce nonzero Christoffel symbols even though the Riemann tensor is zero.

For example, the Euclidean plane in polar coordinates has metric

$$ds^2 = dr^2 + r^2 d\theta^2.$$

The Christoffel symbols include

$$\Gamma^r{}_{\theta\theta} = -r, \qquad \Gamma^\theta{}_{r\theta} = \Gamma^\theta{}_{\theta r} = \frac{1}{r}.$$

These do not indicate intrinsic curvature. The Riemann tensor still vanishes.

In general relativity this distinction is essential. A gravitational field is not identified with Christoffel symbols alone, because they can be made to vanish at a point by choosing local inertial coordinates. Curvature, not the connection coefficients by themselves, encodes invariant gravitational tidal effects.

3.36 Flat Spacetime as a Lorentzian Manifold

Special relativity is described by Minkowski spacetime. As a manifold it is $\mathbb{R}^4$, equipped with the Lorentzian metric

$$\eta = -c^2 dt^2 + dx^2 + dy^2 + dz^2.$$

In units $c = 1$,

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2.$$

In standard inertial coordinates, the metric components are constant, so the Christoffel symbols vanish:

$$\Gamma^{\rho}{}_{\mu\nu} = 0.$$

Consequently,

$$R^{\rho}{}_{\sigma\mu\nu} = 0.$$

However, if one uses accelerated or curvilinear coordinates on Minkowski spacetime, Christoffel symbols may become nonzero. The curvature remains zero. Thus special relativity is not the absence of all connection coefficients in all coordinates; it is the vanishing of the Riemann curvature tensor of the Minkowski metric.

3.37 Curved Spacetime and Gravity

In general relativity, the gravitational field is represented by a Lorentzian metric g on a four-dimensional smooth manifold M. The metric determines:

1. the causal cones;
2. proper time along timelike curves;
3. spatial distances relative to observers;
4. the Levi-Civita connection;
5. the geodesics of free particles and light rays;
6. the curvature tensors entering the field equations.

The geometry is dynamical: the metric is not fixed in advance. It is solved from Einstein's equation together with matter equations and appropriate initial or boundary conditions.

The statement that gravity is curvature should be read carefully. The connection controls free-fall equations, while curvature controls invariant tidal effects. The metric determines both. Standard general relativity packages these facts in the geometry of a Lorentzian manifold.

3.38 Initial Data and the Role of Hypersurfaces

A spacetime can often be studied by splitting it into space plus time. A spacelike hypersurface $\Sigma \subset M$ has an induced Riemannian metric h and extrinsic curvature K. The pair (h, K) forms the geometric part of initial data.

The Einstein equation imposes constraint equations on this data. In vacuum, these are schematically

$$^{(3)}R + K^2 - K_{ij}K^{ij} = 0,$$

and

$$D_j(K^{ij} - h^{ij}K) = 0,$$

where $^{(3)}R$ is the scalar curvature of the spatial metric h, and D is its Levi-Civita connection.

This shows that general relativity is not merely a set of four-dimensional formulae. It also has an initial value formulation: one may specify suitable data on a spatial hypersurface and evolve them, subject to constraints.

This section only indicates the structure. A rigorous treatment requires global hyperbolicity, Sobolev spaces, and the analysis of nonlinear hyperbolic systems.

3.39 Summary of the Mathematical Pipeline

The mathematical construction used in general relativity can be summarized as follows.

1. A smooth four-dimensional manifold M represents spacetime as a set of events with differentiable structure.
2. Tangent spaces T_pM contain possible infinitesimal directions at each event.
3. Tensor fields provide coordinate-independent physical fields.
4. A Lorentzian metric g classifies directions as timelike, null, or spacelike and determines proper time.
5. The metric determines a unique Levi-Civita connection ∇.
6. The connection defines parallel transport and geodesics.
7. The Riemann tensor measures curvature as noncommutation of covariant derivatives and path dependence of parallel transport.
8. The Ricci tensor and scalar curvature are contractions of the Riemann tensor.
9. The Einstein tensor is the divergence-free curvature tensor that couples to stress-energy.

Summary

A spacetime model in standard general relativity is a pair (M, g), where M is a smooth four-dimensional manifold and g is a Lorentzian metric. The Levi-Civita connection of g defines free fall; its curvature defines invariant tidal gravity; contractions of that curvature form the Einstein tensor

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu},$$

which appears in the field equation

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}.$$

Chapter 4

Special Relativity: Mathematical Structure and Physical Consequences

4.1 Aim and Scope

This chapter is a self-contained lecture note on special relativity. It is written without graphical TikZ input files. Places where a spacetime diagram would be pedagogically useful are marked by explicit placeholders. The purpose is to establish a reliable mathematical and conceptual text first; graphical material can later be added one diagram at a time after separate compilation checks.

Special relativity is a theory of spacetime structure. Its most important message is not only that moving clocks and rods behave in unfamiliar ways. The deeper statement is that space and time coordinates are frame-dependent, while a certain quadratic spacetime interval is invariant. Once this invariant structure is accepted, the familiar results – relativity of simultaneity, time dilation, length contraction, velocity addition, and relativistic energy and momentum – become consequences of one coherent geometry.

Throughout the chapter we use the following notation. An inertial frame is denoted by S, another frame moving with constant velocity v along the positive x-axis is denoted by S', and

$$\beta = \frac{v}{c}, \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}}.$$

Here c is the speed of light in vacuum. We use coordinates

$$(ct, x, y, z),$$

so that the time coordinate has the same physical dimension as the spatial coordinates. Unless explicitly stated otherwise, the metric convention is

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2.$$

Remark

This chapter deliberately contains no active spacetime diagrams. Figure slots are written as ordinary boxed text rather than TikZ code. This keeps the build stable while the mathematical content is developed.

4.2 Events, Coordinates, and Inertial Frames

An *event* is something localized in space and time. Examples include a flash of light, the emission of a particle, the arrival of a signal, or one tick of a clock. In an inertial frame S, an event is assigned four coordinates

$$(c\Delta t, \Delta x, \Delta y, \Delta z),$$

but a single event itself is normally written as

$$(ct, x, y, z).$$

The same physical event may have different coordinates in a different inertial frame S'. The transformation between inertial coordinates is the main mathematical object of special relativity.

An inertial frame is a frame in which a free particle moves with constant velocity. Equivalently, it is a frame without acceleration or rotation. The principle of relativity states that no inertial frame is physically preferred. The laws of physics must have the same form in all inertial frames.

For most derivations it is enough to use the standard configuration:

- the axes of S and S' are parallel,
- S' moves with velocity v along the x-axis of S,
- the origins coincide at $t = t' = 0$,
- transverse coordinates are unchanged: $y' = y$, $z' = z$.

Then the nontrivial part of the theory is the relation between (ct, x) and (ct', x').

Definition

A *worldline* is the set of events occupied by an object as time evolves. In a spacetime diagram with vertical coordinate ct and horizontal coordinate x, a particle at rest in frame S has a vertical worldline. A particle moving with constant velocity has a straight tilted worldline.

Placeholder for a later spacetime diagram: events, worldlines, and inertial axes.

4.3 Postulates of Special Relativity

Special relativity is based on two physical postulates.

1. **Principle of relativity.** The laws of physics have the same form in all inertial frames.
2. **Invariance of the speed of light.** Light in vacuum propagates with the same speed c in every inertial frame, independently of the motion of the source or the observer.

The second postulate is incompatible with Galilean transformations. In Newtonian kinematics one writes

$$x' = x - vt, \qquad t' = t.$$

If a light ray satisfies $x = ct$ in frame S, then the Galilean formulas give

$$x' = x - vt = (c - v)t, \qquad t' = t,$$

and therefore

$$\frac{x'}{t'} = c - v.$$

The light speed would depend on the observer's motion. This contradicts the second postulate. The transformation law between inertial frames must therefore be different.

A crucial assumption, usually left implicit, is that transformations between inertial frames are linear. Linearity follows from homogeneity of space and time: the same experiment performed at another place or another time should not change the transformation rule. It also ensures that uniform motion is mapped to uniform motion.

Remark

The postulates do not say that all observers see the same coordinates. They say that the laws of physics have the same form and that the speed c is invariant. Coordinate time and coordinate distance become frame-dependent quantities.

4.4 The Spacetime Interval

The central invariant of special relativity is the spacetime interval. For two events with coordinate differences

$$(c\Delta t, \Delta x, \Delta y, \Delta z),$$

define

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2.$$

This quantity is invariant under Lorentz transformations. In other words, if the same two events are described in another inertial frame S', then

$$\Delta s'^2 = \Delta s^2.$$

The interval classifies separations between events.

Type	Condition	Physical meaning
Timelike	$\Delta s^2 > 0$	A massive particle can connect the events.
Lightlike	$\Delta s^2 = 0$	A light signal can connect the events.
Spacelike	$\Delta s^2 < 0$	No causal signal with speed $\leq c$ can connect them.

For timelike separation, the *proper time* $\Delta\tau$ is defined by

$$\Delta s^2 = c^2\Delta\tau^2.$$

Hence

$$\Delta\tau = \sqrt{\Delta t^2 - \frac{\Delta x^2 + \Delta y^2 + \Delta z^2}{c^2}}.$$

This is the time measured by an ideal clock that travels from the first event to the second event along the corresponding worldline.

For spacelike separation, one may define a proper distance $\Delta\ell$ by

$$\Delta\ell^2 = -\Delta s^2.$$

This is the spatial distance between the events in a frame where the two events are simultaneous.

Placeholder for a later light-cone diagram: timelike, lightlike, and spacelike regions.

4.5 Derivation of the Lorentz Transformation

We now derive the Lorentz transformation in one spatial dimension. Because the origin of S' moves according to $x = vt$ in frame S, the coordinate x' must vanish when $x = vt$. Therefore linearity suggests

$$x' = A(x - vt),$$

where A may depend on v, but not on x or t.

Similarly, by symmetry, the inverse transformation must have the same form with v replaced by $-v$:

$$x = A(x' + vt').$$

The transformation must preserve the light rays $x = ct$ and $x = -ct$. The standard Lorentz transformation satisfying these requirements is

$$x' = \gamma(x - vt),$$

$$t' = \gamma\left(t - \frac{vx}{c^2}\right).$$

In terms of ct, it becomes

$$x' = \gamma(x - \beta ct), \qquad ct' = \gamma(ct - \beta x).$$

The inverse transformation is

$$x = \gamma(x' + vt'),$$

$$t = \gamma\left(t' + \frac{vx'}{c^2}\right),$$

or equivalently

$$x = \gamma(x' + \beta ct'), \qquad ct = \gamma(ct' + \beta x').$$

Let us verify interval invariance in one spatial dimension. We compute

$$(ct')^2 - (x')^2 = \gamma^2\left[(ct - \beta x)^2 - (x - \beta ct)^2\right].$$

Expanding the bracket gives

$$(ct - \beta x)^2 - (x - \beta ct)^2 = c^2t^2 - 2\beta ctx + \beta^2x^2 - x^2 + 2\beta ctx - \beta^2c^2t^2.$$

The mixed terms cancel, so

$$(ct - \beta x)^2 - (x - \beta ct)^2 = (1 - \beta^2)c^2t^2 - (1 - \beta^2)x^2.$$

Thus

$$(ct')^2 - (x')^2 = \gamma^2(1 - \beta^2)(c^2t^2 - x^2).$$

Since

$$\gamma^2(1 - \beta^2) = 1,$$

we obtain

$$(ct')^2 - (x')^2 = c^2t^2 - x^2.$$

This proves the invariance of the interval in the (ct, x)-plane.

4.6 Matrix Form and the Metric

Introduce the two-component spacetime column vector

$$X = \begin{pmatrix} ct \\ x \end{pmatrix}.$$

A Lorentz boost along the x-axis can be written as

$$X' = \Lambda X,$$

where

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix}.$$

The interval in one spatial dimension is

$$\Delta s^2 = X^T \eta X, \qquad \eta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The invariance condition is

$$\Lambda^T \eta \Lambda = \eta.$$

This equation is the analogue of the Euclidean rotation condition

$$R^T I R = I,$$

but with the Minkowski metric η instead of the Euclidean identity matrix.

In four-dimensional notation one writes

$$x^\mu = (ct, x, y, z),$$

with Greek indices $\mu, \nu = 0, 1, 2, 3$. With the chosen sign convention,

$$\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1).$$

The interval is

$$\Delta s^2 = \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu.$$

The Lorentz transformation satisfies

$$\eta_{\rho\sigma} \Lambda^\rho{}_\mu \Lambda^\sigma{}_\nu = \eta_{\mu\nu}.$$

This compact expression is the algebraic form of spacetime geometry.

4.7 Rapidity and Hyperbolic Rotations

The Lorentz transformation resembles a rotation, but it preserves $(ct)^2 - x^2$ instead of $x^2 + y^2$. This is why hyperbolic functions appear. Define the rapidity φ by

$$\tanh\varphi = \beta = \frac{v}{c}.$$

Then

$$\cosh\varphi = \gamma, \qquad \sinh\varphi = \gamma\beta.$$

The Lorentz transformation becomes

$$ct' = ct\cosh\varphi - x\sinh\varphi,$$

$$x' = x\cosh\varphi - ct\sinh\varphi.$$

This is a hyperbolic rotation in the (ct, x)-plane. It preserves

$$(ct')^2 - (x')^2 = (ct)^2 - x^2.$$

The advantage of rapidity is that rapidities add. If one boost has rapidity φ_1, and a second boost in the same direction has rapidity φ_2, then the combined boost has rapidity

$$\varphi = \varphi_1 + \varphi_2.$$

Velocity itself does not add linearly; rapidity does.

Placeholder for a later diagram: Lorentz axes as hyperbolic rotations toward the light cone.

4.8 Relativity of Simultaneity

The time transformation is

$$t' = \gamma \left(t - \frac{vx}{c^2} \right).$$

For two events, subtract the equations:

$$\Delta t' = \gamma \left(\Delta t - \frac{v\Delta x}{c^2} \right).$$

Suppose that the two events are simultaneous in frame S. Then $\Delta t = 0$, and therefore

$$\Delta t' = -\gamma \frac{v\Delta x}{c^2}.$$

If $v \neq 0$ and $\Delta x \neq 0$, then $\Delta t' \neq 0$. Thus two spatially separated events that are simultaneous in one inertial frame are not generally simultaneous in another inertial frame.

This is not an optical delay effect. It remains after one corrects for signal travel time. It is a statement about the coordinate assignment used by inertial frames.

Example

Consider two lightning strikes hitting the front and rear of a moving train. A platform observer may judge the strikes simultaneous. The train observer uses a different simultaneity slice and assigns different times to the same two events. Both descriptions are consistent because simultaneity at a distance is not absolute.

Placeholder for a later diagram: two different simultaneity slices through the same spacetime.

4.9 Time Dilation

Time dilation concerns two events occurring at the same place in the rest frame of a clock. Suppose the clock is at rest in S'. Then two ticks of the clock satisfy

$$\Delta x' = 0.$$

Use the inverse Lorentz transformation for time:

$$\Delta t = \gamma \left(\Delta t' + \frac{v \Delta x'}{c^2} \right).$$

Since $\Delta x' = 0$, this reduces to

$$\Delta t = \gamma \Delta t'.$$

The time measured by the clock in its own rest frame is the proper time $\Delta\tau$, so

$$\Delta t' = \Delta\tau.$$

Therefore

$$\boxed{\Delta t = \gamma \Delta\tau.}$$

Because $\gamma \geq 1$, the coordinate time measured in the frame where the clock is moving is larger than the proper time measured by the clock itself. This is the statement that a moving clock is measured to run slow.

The same result follows from the interval. For a clock moving with speed u in frame S, we have

$$\Delta x = u \Delta t.$$

Then

$$c^2 \Delta\tau^2 = c^2 \Delta t^2 - \Delta x^2 = c^2 \Delta t^2 - u^2 \Delta t^2 = c^2 \Delta t^2 \left(1 - \frac{u^2}{c^2} \right).$$

Thus

$$\Delta\tau = \Delta t \sqrt{1 - \frac{u^2}{c^2}} = \frac{\Delta t}{\gamma_u}.$$

Equivalently,

$$\Delta t = \gamma_u \Delta\tau.$$

Placeholder for a later diagram: two ticks on one moving clock worldline.

4.10 Length Contraction

Length contraction concerns a rod at rest in frame S'. Let its endpoints have fixed coordinates x'_1 and x'_2, so its proper length is

$$L_0 = x'_2 - x'_1.$$

To measure the rod's length in frame S, one must record the positions of the two endpoints at the same time in S. Hence the two measurement events satisfy

$$\Delta t = 0.$$

The spatial Lorentz transformation gives

$$\Delta x' = \gamma(\Delta x - v\Delta t).$$

With $\Delta t = 0$, this becomes

$$\Delta x' = \gamma \Delta x.$$

Since $\Delta x' = L_0$ and $\Delta x = L$, we get

$$L_0 = \gamma L.$$

Therefore

$$\boxed{L = \frac{L_0}{\gamma}.}$$

A moving rod is shorter along the direction of motion than its proper length. Lengths perpendicular to the direction of motion are unchanged by a boost in the x-direction.

The simultaneity condition is essential. If the two endpoints are not measured at the same time in the measuring frame, the result is not a length measurement in that frame.

Remark

Length contraction is not mechanical compression. The rod is not squeezed in its own rest frame. It has length L_0 there. A different frame measures a shorter length because it uses a different simultaneity condition for comparing endpoint events.

Placeholder for a later diagram: endpoint worldlines and simultaneous measurement slices.

4.11 Velocity Addition

Let an object move with velocity $u' = dx'/dt'$ in frame S'. Frame S' moves with velocity v relative to frame S. From the inverse Lorentz transformation,

$$dx = \gamma(dx' + vdt'),$$

$$dt = \gamma\left(dt' + \frac{v}{c^2}dx'\right).$$

Therefore

$$u = \frac{dx}{dt} = \frac{dx' + vdt'}{dt' + (v/c^2)dx'}.$$

Dividing numerator and denominator by dt' gives

$$\boxed{u = \frac{u' + v}{1 + u'v/c^2}.}$$

This formula replaces the Galilean rule $u = u' + v$.

If $u' = c$, then

$$u = \frac{c + v}{1 + v/c} = c.$$

Thus light has speed c in both frames. If $|u'| < c$ and $|v| < c$, then $|u| < c$. Relativistic velocity addition never pushes a subluminal speed past c.

For three-dimensional motion with a boost along x, the components transform as

$$u_x = \frac{u'_x + v}{1 + u'_x v/c^2},$$

$$u_y = \frac{u'_y}{\gamma(1 + u'_x v/c^2)}, \qquad u_z = \frac{u'_z}{\gamma(1 + u'_x v/c^2)}.$$

The transverse components are affected because time transforms.

4.12 Proper Time and Worldline Length

For a general worldline $\mathbf{x}(t)$, define the ordinary speed

$$u(t) = \left| \frac{d\mathbf{x}}{dt} \right| .$$

The proper time increment is obtained from the interval:

$$c^2 d\tau^2 = c^2 dt^2 - d\mathbf{x}^2 .$$

Since

$$d\mathbf{x}^2 = u(t)^2 dt^2,$$

we find

$$d\tau = dt\sqrt{1 - \frac{u(t)^2}{c^2}} .$$

The total proper time accumulated along the worldline is therefore

$$\tau = \int d\tau = \int_{t_1}^{t_2} \sqrt{1 - \frac{u(t)^2}{c^2}}\, dt .$$

This formula gives the elapsed time on an ideal clock carried along the worldline.

For constant speed u, it reduces to

$$\Delta\tau = \Delta t\sqrt{1 - \frac{u^2}{c^2}} = \frac{\Delta t}{\gamma_u} .$$

The twin paradox is resolved by this formula: different worldlines connecting the same two events can have different accumulated proper times. The traveling twin follows a non-inertial path if departure and reunion occur at the same place, and the proper time integral differs from that of the stay-at-home twin.

4.13 Four-Velocity and Four-Momentum

The position four-vector is

$$x^\mu = (ct, x, y, z).$$

The proper time allows us to define the four-velocity

$$U^\mu = \frac{dx^\mu}{d\tau}.$$

Because

$$\frac{dt}{d\tau} = \gamma_u,$$

we obtain

$$U^\mu = \gamma_u(c, u_x, u_y, u_z).$$

Its invariant norm is

$$U^\mu U_\mu = c^2.$$

This is the relativistic replacement for the Newtonian velocity vector as the geometrically natural velocity object.

The four-momentum of a particle with rest mass m is

$$p^\mu = mU^\mu.$$

Therefore

$$p^\mu = (\gamma mc, \gamma m\mathbf{u}).$$

Writing

$$p^\mu = \left(\frac{E}{c}, \mathbf{p}\right),$$

we identify

$$E = \gamma mc^2, \qquad \mathbf{p} = \gamma m\mathbf{u}.$$

The invariant norm of four-momentum is

$$p^\mu p_\mu = m^2c^2.$$

In terms of energy and momentum,

$$\frac{E^2}{c^2} - p^2 = m^2c^2.$$

Multiplying by c^2 gives the energy-momentum relation

$$\boxed{E^2 = p^2c^2 + m^2c^4.}$$

For a particle at rest, $p = 0$, hence

$$\boxed{E_0 = mc^2.}$$

For a massless particle, $m = 0$, hence

$$E = pc.$$

4.14 Relativistic Energy at Low Velocity

The relativistic energy is

$$E = \gamma mc^2.$$

For $u \ll c$, write

$$\gamma = \frac{1}{\sqrt{1 - u^2/c^2}}.$$

Using the binomial approximation

$$(1 - \epsilon)^{-1/2} \approx 1 + \frac{1}{2}\epsilon \qquad (\epsilon \ll 1),$$

we obtain

$$\gamma \approx 1 + \frac{1}{2}\frac{u^2}{c^2}.$$

Therefore

$$E = \gamma mc^2 \approx mc^2 + \frac{1}{2}mu^2.$$

The first term is the rest energy, and the second term is the Newtonian kinetic energy. Thus special relativity contains Newtonian mechanics as a low-speed approximation.

The relativistic kinetic energy is defined as

$$K = E - mc^2 = (\gamma - 1)mc^2.$$

At low speeds,

$$K \approx \frac{1}{2}mu^2.$$

At speeds close to c, γ grows without bound, so the kinetic energy needed to accelerate a massive particle toward c grows without bound. This is why massive particles cannot be accelerated to the speed of light.

4.15 Muon Lifetime Example

Cosmic rays striking the upper atmosphere produce muons. A muon at rest has a mean lifetime approximately

$$\tau_0 \approx 2.2\,\mu\text{s}.$$

Classically, even if a muon moved almost at the speed of light, it would travel roughly

$$c\tau_0 \approx (3.0 \times 10^8\,\text{m/s})(2.2 \times 10^{-6}\,\text{s}) \approx 660\,\text{m}.$$

Many muons are produced several kilometres above Earth's surface, yet many are detected near the ground. Special relativity explains this.

In the Earth frame, the muon moves rapidly, so its lifetime is time-dilated:

$$\Delta t = \gamma\tau_0.$$

If $\gamma = 10$, then

$$\Delta t \approx 22\,\mu\text{s}.$$

The distance traveled can then be of order

$$L \approx c\Delta t \approx 6.6\,\text{km}.$$

This is compatible with atmospheric distances.

In the muon rest frame, the muon lifetime is simply τ_0. Instead, the atmosphere moves toward the muon and is length-contracted:

$$L = \frac{L_0}{\gamma}.$$

The two descriptions use different coordinates, but they agree on the physical fact: the muon can reach the detector.

Placeholder for a later diagram: muon worldline and atmospheric distance in two frames.

4.16 Causal Structure

The sign of the interval determines causal relationships. If two events are timelike separated, then there exists an inertial frame in which they occur at the same spatial point. Their time order is the same in every inertial frame. A massive particle can travel from one event to the other if the second lies in the future light cone of the first.

If two events are lightlike separated, only a signal moving at speed c can connect them. Their interval is zero:

$$\Delta s^2 = 0.$$

If two events are spacelike separated, no signal moving at or below c can connect them. Different inertial frames may disagree about their temporal order, but this does not violate causality because neither event can influence the other.

This is the relativistic replacement for Newtonian absolute time. Absolute simultaneity disappears, but causal order for physically connectable events is preserved.

4.17 Common Misinterpretations

4.17.1 Moving clocks do not malfunction

Time dilation is not a statement about defective clocks. It applies to all ideal clocks: atomic clocks, particle lifetimes, mechanical clocks, and biological aging. The reason is geometric: clocks measure proper time along their worldlines.

4.17.2 Length contraction is not compression

A moving rod is not physically squeezed in its own rest frame. It has its proper length L_0 there. A different frame measures a shorter length because it uses a different simultaneity condition for comparing endpoint events.

4.17.3 Relativity of simultaneity is not an optical illusion

Observers may visually see events after light travel delays. Relativity of simultaneity is a deeper statement: after correcting for signal travel time, different inertial frames still assign different time coordinates to separated events.

4.17.4 Nothing massive reaches the speed of light

The speed c is not merely the speed of light. It is the invariant speed of spacetime. Massive particles have timelike worldlines and speeds below c. Massless particles in vacuum move at c.

4.18 Formula Summary

The Lorentz factor is

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

The Lorentz transformation is

$$x' = \gamma(x - vt), \qquad t' = \gamma\left(t - \frac{vx}{c^2}\right).$$

The inverse transformation is

$$x = \gamma(x' + vt'), \qquad t = \gamma\left(t' + \frac{vx'}{c^2}\right).$$

The invariant interval is

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2.$$

The proper time for timelike separation is

$$\Delta\tau = \sqrt{\Delta t^2 - \frac{\Delta x^2 + \Delta y^2 + \Delta z^2}{c^2}}.$$

Time dilation is

$$\Delta t = \gamma\Delta\tau.$$

Length contraction is

$$L = \frac{L_0}{\gamma}.$$

Relativity of simultaneity is expressed by

$$\Delta t' = \gamma\left(\Delta t - \frac{v\Delta x}{c^2}\right).$$

Velocity addition in one dimension is

$$u = \frac{u' + v}{1 + u'v/c^2}.$$

Relativistic momentum and energy are

$$\mathbf{p} = \gamma m\mathbf{u}, \qquad E = \gamma mc^2.$$

The invariant energy-momentum relation is

$$E^2 = p^2c^2 + m^2c^4.$$

Summary

Special relativity is the geometry of Minkowski spacetime. Lorentz transformations preserve the spacetime interval. Time dilation, length contraction, relativity of simultaneity, velocity addition, and relativistic energy-momentum are not separate tricks; they are consequences of the same invariant spacetime structure.

Chapter 5

Special Relativity: Spacetime Diagram Gallery

This chapter collects the first locally tested spacetime diagrams for the special relativity note. The diagrams are intentionally simple and robust. They are included as TikZ inputs from the project figure directory and can later be refined one by one.

5.1 Light Cone and Interval Classification

The light cone separates timelike, lightlike, and spacelike directions. With our sign convention,

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2,$$

timelike separations have $\Delta s^2 > 0$, lightlike separations have $\Delta s^2 = 0$, and spacelike separations have $\Delta s^2 < 0$.

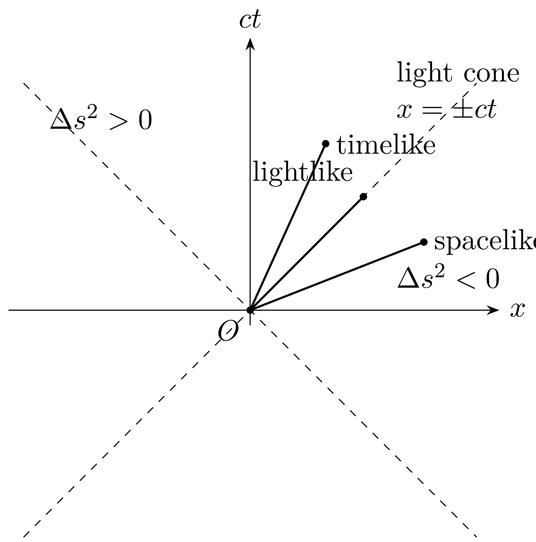

Figure 5.1: The light cone and the classification of spacetime intervals.

5.2 Lorentz Axes

A Lorentz boost changes both the time and space axes. The transformed axes tilt toward the light cone. Rapidity φ is defined by

$$\tanh \varphi = \beta = \frac{v}{c}.$$

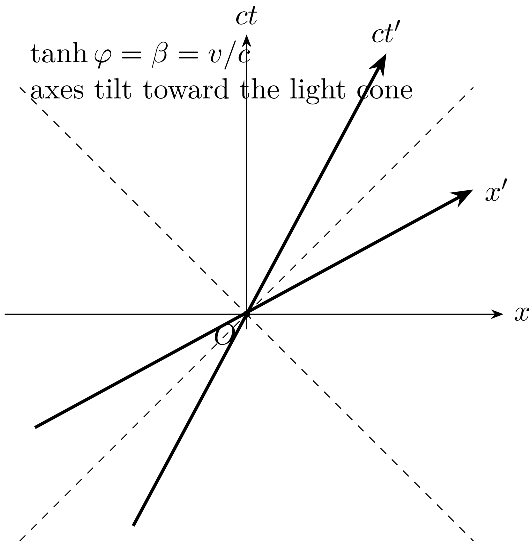

Figure 5.2: Lorentz-transformed axes in a spacetime diagram.

5.3 Relativity of Simultaneity

The transformation

$$\Delta t' = \gamma \left(\Delta t - \frac{v \Delta x}{c^2} \right)$$

shows that events simultaneous in one frame need not be simultaneous in another frame. In spacetime geometry, this is represented by different simultaneity slices.

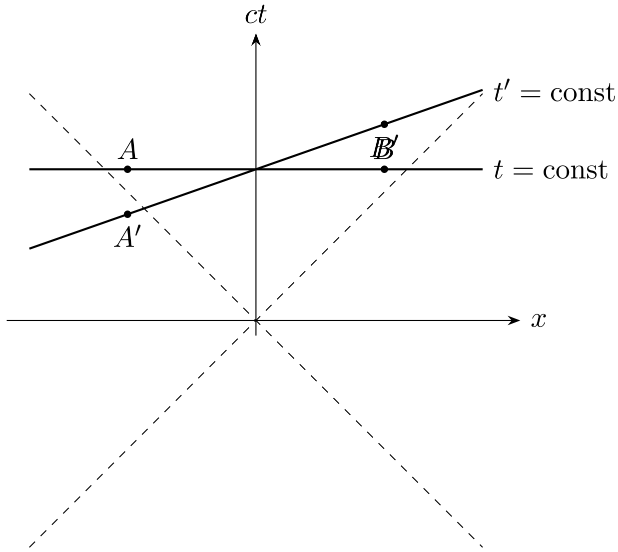

Figure 5.3: Two different simultaneity slices for observers in relative motion.

5.4 Time Dilation

Two ticks of one moving clock lie on the same worldline. The proper time is the invariant time measured along that clock's worldline:

$$(c\Delta\tau)^2 = (c\Delta t)^2 - (\Delta x)^2.$$

This gives

$$\Delta t = \gamma\Delta\tau.$$

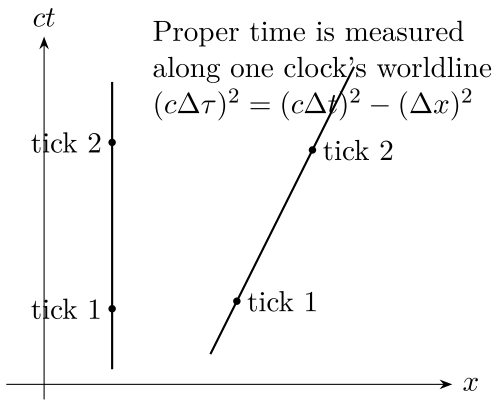

Figure 5.4: Time dilation represented by two ticks on a clock worldline.

5.5 Length Contraction

A length measurement compares the positions of both endpoints at the same time in the measuring frame. The resulting contracted length is

$$L = \frac{L_0}{\gamma}.$$

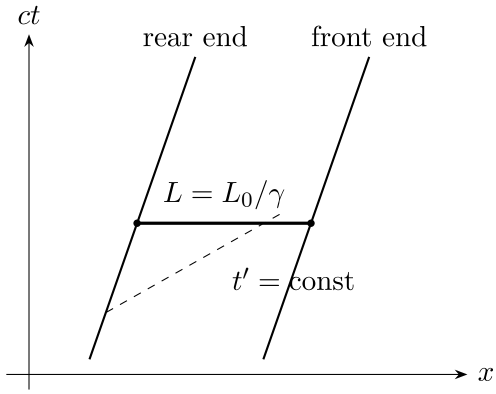

Figure 5.5: Length contraction from simultaneous endpoint measurements.

5.6 Muon Lifetime Geometry

Muon survival can be described in two equivalent ways. In the Earth frame, the muon's lifetime is dilated. In the muon frame, the atmospheric distance is length-contracted. Both descriptions refer to the same pair of physical events.

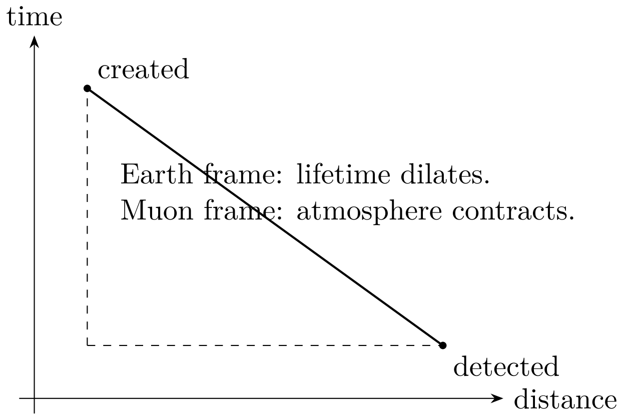

Figure 5.6: Muon survival described by time dilation or length contraction.